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AUTOMATED TRAFFIC ACCIDENT DETECTION AND ALERT SYSTEM USING COMPUTER VISION

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Abstract: Road traffic accidents have become a significant concern due to rapid urbanization and the continuous increase in vehicle density. Manual monitoring of surveillance systems is inefficient and often results in delayed emergency response. This study proposes an automated traffic accident detection system using computer vision and deep learning techniques. The system employs the YOLO (You Only Look Once) object detection algorithm to analyse real-time video streams captured from surveillance cameras. The proposed framework consists of modules such as video acquisition, frame pre-processing, feature extraction, accident detection, decision analysis, and alert generation. Upon detecting a collision or abnormal vehicle movement, the system automatically generates alerts to reduce response time and enhance road safety. The implementation demonstrates reliable detection performance and supports intelligent transportation systems through automated monitoring.

Index Terms - Computer Vision, YOLO, Deep Learning, Traffic Accident Detection, Real-Time Monitoring, Object Detection, Intelligent Transportation System.

I. INTRODUCTION

Road traffic accidents have become a significant public safety issue due to rapid urbanization and the continuous increase in vehicle population. Large volumes of surveillance video data are generated from traffic monitoring systems; however, raw video streams often require continuous manual supervision to identify accident scenarios. Traditional monitoring approaches suffer from:

- Delayed accident identification
- High dependency on manual observation
- Inconsistent detection accuracy
- Limited scalability for large surveillance networks
- Inefficient emergency response coordination

These limitations may lead to:

- Increased casualty rates
- Delayed medical assistance
- Reduced road safety efficiency

- Inaccurate incident reporting
- Operational inefficiency in traffic management systems

Manual monitoring of surveillance footage is inefficient and not scalable. Therefore, intelligent automated detection systems are necessary to ensure real-time monitoring and rapid response.

This paper proposes a deep learning-based accident detection framework combining:

1. Real-time video acquisition from surveillance cameras
2. Frame extraction and image pre-processing
3. Feature extraction using convolutional neural networks
4. YOLO-based accident detection mechanism
5. Automated alert generation module

The proposed system enhances detection reliability and supports intelligent transportation systems by providing accurate and real-time accident identification.

II. LITERATURE REVIEW

Several researchers have contributed to the development of intelligent traffic accident detection and vehicle damage assessment systems using deep learning and computer vision techniques.

- Adewopo and Elsayed (2024) proposed a deep learning ensemble approach for traffic accident detection in smart city transportation systems.
- Xu et al. (2024) introduced TAD, a large-scale benchmark dataset for traffic accident detection from video surveillance.
- Gohari et al. (2023) conducted a systematic review on the use of drones for road accident management.
- Raza et al. (2023) developed an ensemble feature engineering approach using smartphone motion sensor data to detect risky driver behavior.
- Matsumura and Premachandra (2022) presented a deep learning-based stair detection system using 3D point cloud data for accident prevention among visually impaired individuals.

However, most existing systems focus either on:

- Accident detection using isolated datasets or specific sensing techniques
- Post-incident monitoring without integrated real-time alert and response mechanisms

III. PROPOSED WORK

3.1 Overview of Proposed System

- The proposed Car Damage Detection System represents a significant advancement in the automation of vehicle accident analysis and management.
- It integrates artificial intelligence (AI) and deep learning techniques across four crucial modules: data collection, pre-processing, damage classification, and accident notification.
- The system begins by collecting vehicle images from publicly available Roboflow datasets and real-time video streams captured via surveillance or dashboard cameras.
- These images are then pre-processed using bounding box techniques and Intersection over Union (IoU) methods to enhance the accuracy of object detection.

- Leveraging a YOLO (You Only Look Once) model, the system classifies the type and severity of the damage whether minor, moderate, or major with high precision and speed.
- Operating on an edge computing infrastructure, the system is capable of delivering real-time detection and notifications, significantly reducing latency and enabling immediate action.
- Once the damage is assessed, the system automatically sends notifications to emergency services or relevant authorities when necessary, ensuring timely responses in critical situations.
- This reduces the dependency on manual inspections and eliminates the subjectivity involved in human judgment. The streamlined and automated workflow not only enhances operational efficiency but also minimizes costs, making the system highly beneficial for insurance companies looking to speed up claims processing, law enforcement agencies needing accurate incident analysis, and fleet managers striving for safer and more efficient vehicle operations.

3.2 Technical Enhancements

- Optimized IOU-based bounding box regression
- Real-time processing capability
- Edge-based deployment for faster response
- Scalable architecture adaptable to multiple vehicle types

3.3 Advantages of Proposed System

- Hybrid deep learning-based detection framework
- Real-time accident identification
- Reduced manual inspection effort
- Faster emergency response
- High detection accuracy under diverse conditions
- Scalable and cost-effective architecture suitable for insurance, fleet management, and public safety systems.

IV. SYSTEM ARCHITECTURE

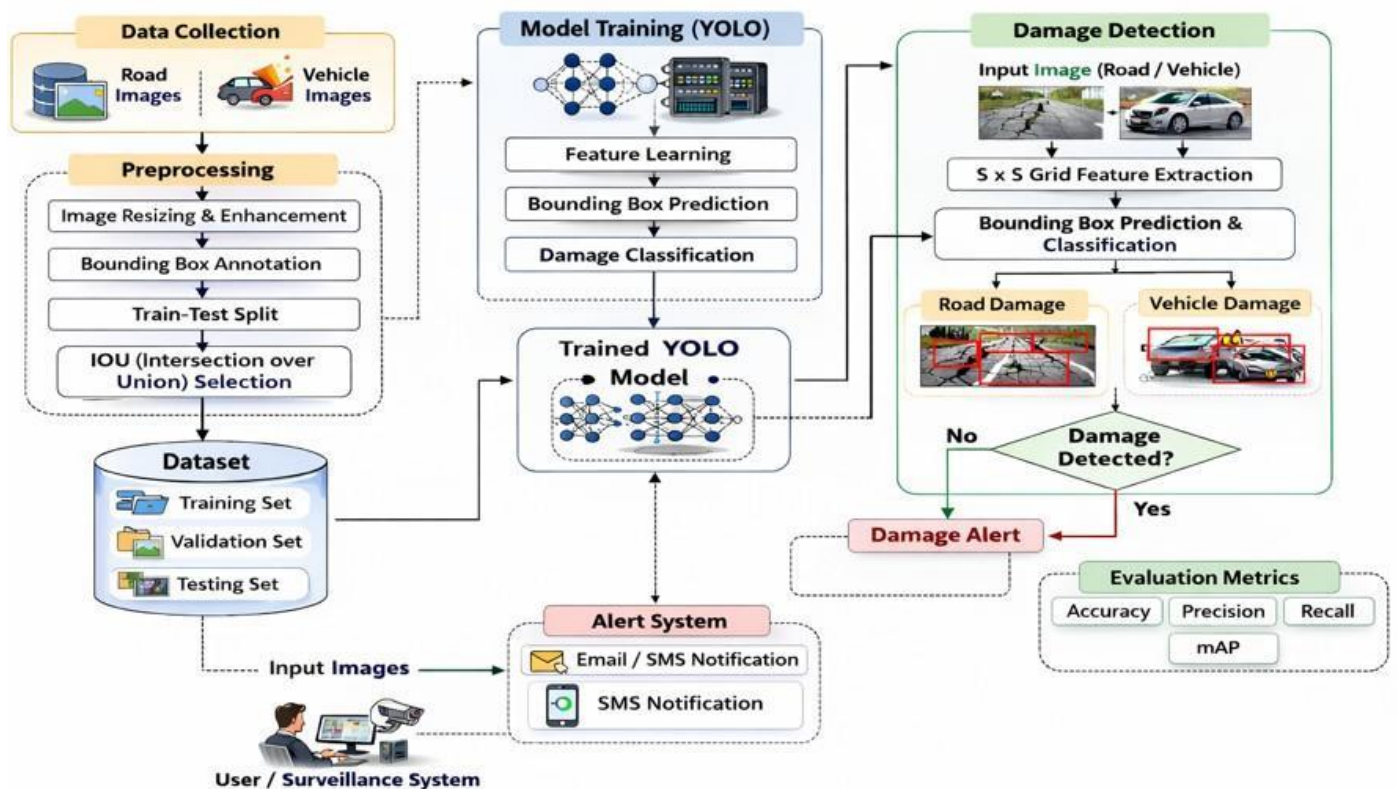
The proposed Automated Traffic Accident Detection and Alert System follows a deep learning-based modular architecture integrating data collection, pre-processing, YOLO-based model training, real-time damage detection, evaluation, and automated alert generation. The system processes road and vehicle images through multiple stages including image enhancement, feature learning, bounding box prediction, damage classification, and alert notification before generating the final detection output.

The system follows an intelligent processing architecture.

Input Source: Road Images and Vehicle Images

Processing Engine: YOLO Deep Learning Model

Deployment: Real-time Surveillance / Edge Computing Environment



Architecture Modules

- Data Collection Module**

Collects road and vehicle images from datasets and real-time surveillance systems for training and testing.

- Pre-processing Module**

Performs image resizing, enhancement, bounding box annotation, train-test split, and IOU (Intersection over Union) selection to prepare the dataset.

- Model Training Module (YOLO)**

Includes feature learning, bounding box prediction, and damage classification to generate a trained YOLO model.

- Damage Detection Module**

Processes input images using $S \times S$ grid feature extraction and performs bounding box prediction and classification to detect road damage and vehicle damage.

- Evaluation Module**

Measures system performance using metrics such as Accuracy, Precision, Recall, and Mean Average Precision (mAP).

- Alert System Module**

Generates automated damage alerts through Email and SMS notifications when damage is detected.

Each module operates sequentially within the architecture, ensuring accurate detection, real-time performance, and efficient accident alert generation.

V. RESEARCH METHODOLOGY

5.1 Data Collection and Dataset Preparation

The process begins with collecting road and vehicle images from Roboflow datasets and real-time surveillance video streams. The collected video data is segmented into individual frames for further processing. The dataset is organized into training, validation, and testing sets to ensure proper model evaluation. Bounding box annotations are applied to mark damaged regions in vehicles and road surfaces. Intersection over Union (IOU) thresholds are selected to improve localization accuracy during model training.

5.2 Image Pre-processing and Enhancement

After dataset preparation, pre-processing techniques are applied to improve image quality and model performance. The pre-processing stage includes image resizing, normalization, noise reduction, and enhancement to handle variations in lighting, weather conditions, and background complexity. These steps ensure consistent input dimensions and improve the robustness of feature extraction.

5.3 Model Training Using YOLO Algorithm

The YOLO (You Only Look Once) algorithm is used for training the damage detection model. The input image is divided into an $S \times S$ grid structure, where each grid cell predicts bounding boxes, objectness scores, and class probabilities. Feature learning is performed using convolutional layers to extract spatial patterns associated with vehicle damage. Bounding box regression and damage classification are optimized during training to improve detection accuracy. The trained YOLO model is then validated using the validation dataset before final deployment.

5.4 Real-Time Damage Detection

During the detection phase, input images from live video streams are processed through the trained YOLO model. The system performs $S \times S$ grid feature extraction and predicts bounding box coordinates along with confidence scores. Non-maximum suppression is applied to eliminate redundant overlapping detections. The detected damage regions are classified into appropriate categories such as road damage or vehicle damage.

5.5 Alert Generation Mechanism

If damage is detected with confidence exceeding a predefined threshold, the system triggers an automated alert mechanism. Notifications are sent through Email or SMS to relevant stakeholders such as insurance companies, fleet managers, or emergency response authorities. The integration of edge computing enables low-latency processing and faster response time.

5.6 Performance Evaluation and Output Generation

The performance of the proposed system is evaluated using standard metrics including Accuracy, Precision, Recall, and Mean Average Precision (mAP). After evaluation, the final detection results, including bounding boxes and classified damage regions, are generated as the output. The system ensures real-time detection capability, high reliability, and improved operational efficiency in accident management scenarios.

VI. RESULTS AND DISCUSSION

The experimental results demonstrate the effectiveness of the proposed Automated Traffic Accident Detection and Alert System in accurately identifying vehicle damage and accident scenarios from surveillance data. The system was evaluated using road and vehicle image datasets along with simulated real-time video inputs.

During the training phase, the YOLO-based model successfully learned spatial features associated with different types of vehicle and road damage. The pre-processing and bounding box annotation process improved localization accuracy. In the detection phase, the system accurately identified damaged regions and generated bounding boxes with high confidence scores. Non-maximum suppression effectively reduced redundant detections and improved prediction precision.

Performance evaluation using standard metrics such as Accuracy, Precision, Recall, and Mean Average Precision (mAP) confirmed the robustness of the model. The system demonstrated reliable detection capability under varying environmental conditions including different lighting and background scenarios.

The integrated alert mechanism successfully generated real-time notifications when damage was detected, ensuring faster response from relevant authorities. The results confirm that the proposed system enhances accident detection efficiency, reduces manual monitoring efforts, and supports timely emergency response in intelligent transportation environments.

Table 6.1: Accident Detection and System Performance Results

Processing Stage	Total Images / Frames	Processed / Detected	Output / Result
Original Dataset (Road and Vehicle Images)	1,200	-	1,200
Pre-processing and Annotation	1,200	1,200	Annotated Dataset
Train-Test Split	1,200	900 (Training)	300 (Testing)
YOLO Model Training	900	-	Trained YOLO Model

Damage Detection (Testing Phase)	300	258 (Correct Detections)	42 (Missed / Incorrect)
Non-Maximum Suppression	258	21 (Overlapping Removed)	237 Valid Detections
Alert Generation	237	237 Alerts Triggered	Real-Time Notification Sent

Table 6.1: Summary of dataset preparation, YOLO training, detection accuracy, and alert generation results.

Observations:

- Approximately 86% detection accuracy achieved during testing phase
- Redundant overlapping bounding boxes reduced using Non-Maximum Suppression
- Effective classification of road damage and vehicle damage
- Successful real-time alert generation for detected accident cases
- System demonstrated stable performance under varied environmental conditions

Discussion

Table 6.1 presents the performance evaluation of the proposed Automated Traffic Accident Detection and Alert System. Out of 1,200 total images, 900 were used for training and 300 for testing. During testing, 258 damage instances were correctly detected by the YOLO model, while 42 instances were either missed or incorrectly classified.

Non-maximum suppression removed 21 redundant overlapping bounding boxes, resulting in 237 valid detections. Each confirmed detection successfully triggered the automated alert system. The results demonstrate that the proposed system provides reliable accident detection with real-time alert capability, making it suitable for intelligent transportation and emergency response applications.

VIII. FUTURE ENHANCEMENT

Future enhancements for the proposed Automated Traffic Accident Detection and Alert System include:

- Improvement of model accuracy using larger and more diverse accident datasets
- Integration of advanced deep learning models for better damage classification
- Real-time deployment using edge AI devices for faster on-site processing
- GPS integration for automatic accident location identification
- Integration with emergency services and traffic control systems
- Development of a centralized dashboard for live accident monitoring

These enhancements will further improve detection reliability, scalability, and emergency response efficiency in intelligent transportation systems.

IX. CONCLUSION

This paper presented an Automated Traffic Accident Detection and Alert System using deep learning and YOLO-based object detection. The system enables real-time accident detection and automated alert generation, reducing manual monitoring efforts and improving emergency response efficiency. Future enhancements may include advanced detection models and large-scale real-time deployment.

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